

1 Analysis 1 by Terence Tao

Started reading on 2026-06-14.

- 2026-06-16
 - § 2.2: Read pp. 26-27.
- 2026-06-15
 - § 2.2: Read pp. 24-26. Tried to prove all the propositions/theorems by myself before reading them.
- 2026-06-14
 - § 2.1: Read the whole section. Focused on the construction of the Peano axioms. Encountered the recursion theorem and sketched out the proof idea.

1.1 My solutions for exercises

I'm practicing proof writing as well as problem solving here, so the writing style can vary from one solution to another.

1.1.1 Exercise 2.2.3

1. (a) $a \geq a$.

Proof. We seek an m such that $a = a + m$. Clearly 0 is such an m , and hence $a \geq a$.

2. (b) If $a \geq b$ and $b \geq c$, then $a \geq c$.

Proof. Let m and n be natural numbers such that $a = b + m$ and $b = c + n$. Then we have $a = (c + n) + m$. By the associative law for addition, $a = c + (n + m)$. Therefore $a \geq c$.

3. (c) If $a \geq b$ and $b \geq a$, then $a = b$.

Proof. Let m and n be natural numbers such that $a = b + m$ and $b = a + n$. Then we have $a = (a + n) + m$ and $a = a + (n + m)$, and so $n + m$ must be zero. Thus $n = m = 0$ by Corollary 2.2.9. Substituting 0 for m into $a = b + m$, we conclude that $a = b$.

- (a) Lemma. If $a + m = a$ then $m = 0$.

Proof. $a + m = a = a + 0$. Using the cancellation law gives $m = 0$.

4. (d) $a \geq b$ iff $a + c \geq b + c$.

Proof.

- (a) (i) If $a \geq b$ then $a + c \geq b + c$.

Let $a = b + m$ for some natural number m . Then $a + c = (b + c) + m$, hence $a + c \geq b + c$.

- (b) (ii) If $a + c \geq b + c$ then $a \geq b$.

Since $a + c \geq b + c$, there exists a natural number m such that $a + c = (b + c) + m$. Then $a + c = (b + m) + c$. By the cancellation law $a = b + m$. Hence $a \geq b$.

5. (e) $a < b$ iff $a++ \leq b$.

Proof. First, suppose $a < b$. Then there exists a natural number m such that

$$b = a + m$$

Since $a < b$ implies $a \neq b$, $m \neq 0$. Thus by Lemma 2.2.10, there is a natural number n such that $n++ = m$. Hence

$$b = a + (n++) = (a + n)++ = (a++) + n$$

Therefore $a++ \leq b$.

Conversely, suppose that $a++ \leq b$. Let m be a natural number such that $b = (a++) + m$. Then

$$b = (a++) + m = (a + m)++ = a + (m++)$$

Suppose, for contradiction, that $a = b$. Then

$$a = a + (m++)$$

and cancelling a gives $m++ = 0$. But that contradicts Axiom 2.3, which says $m++ \neq 0$. Thus $a \neq b$.

Since $b = a + (m++)$, we conclude that $a < b$.

6. (f) $a < b$ iff $b = a + d$ for some positive number d .

Proof. First, assume that $a < b$. Then $b = a + d$ for some natural number d . Suppose, for contradiction, that $d = 0$. then $b = a + 0 = a$, contradicting the fact that $a \neq b$. Hence $d \neq 0$, and we conclude d is positive.

Conversely, assume that $b = a + d$ for some positive number d . If $a = b$, $d = 0$ by the same argument used in (e), contradicting the fact that $d \neq 0$. Thus $a \neq b$ and we conclude that $a < b$.